Linear Algebra with Applications in Automatic Control

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Fall Semester

Chapter 4 Linear transformation in unitary spaces

Z. Song

- *The first part-the first 5 sections*

  •1. Introduce the notion: Euclidean space-a linear space with another operation-inner product defined; unitary space, a Euclidean space over the field R

  •2. Characteristics:

  •3. Define distance between vectors, relation between vectors using inner product concept

  •4. Special linear transformations: unitary, normal transformations

- *The second part-the last 5 sections*

  •1. Introduce Hermite matrix, what are the special properties, usefulness, applications,

  •2. Definite matrix, and applications

Chapter 4 Linear transformation in unitary spaces

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- *This is an interesting chapter*

•In mathematics, Euclidean space is the Euclidean plane, the three-dimensional space of Euclidean geometry, and generalizations of these ideas to higher dimensions.

•Euclidean space is defined using Cartesian coordinates and the ideas of analytic geometry. This brings the tools of algebra and calculus to bear on questions of geometry, and has the advantage that it generalizes easily to Euclidean spaces of more than three dimensions.

•From the modern viewpoint, there is essentially only one Euclidean space of each dimension. In dimension one this is the real line; in dimension two it is the Cartesian plane; and in higher dimensions it is a coordinate space with three or more real number coordinates – in short, an *n*-dimensional real coordinate space. A point in Euclidean space may be identified by a tuple of real numbers, and distances are defined using the Euclidean distance formula.

Chapter 4 Linear transformation in unitary spaces

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- *An intuitive overview-2 dimension*
  - One way to think of the Euclidean plane is as a set of points satisfying certain relationships, expressible in terms of distance and angle. For example, there are two fundamental operations on the plane. One is translation, which means a shifting of the plane so that every point is shifted in the same direction and by the same distance. The other is rotation about a fixed point in the plane, in which every point in the plane turns about that fixed point through the same angle

- **Precisely, Euclidean plane as a two-dimensional real vector space equipped with an inner product. For then:**
- the vectors in the vector space correspond to the points of the Euclidean plane,
- the addition operation in the vector space corresponds to translation, and
- the inner product implies notions of angle and distance, which can be used to define rotation.
- **Once the Euclidean plane has been described in this language, it is actually a simple matter to extend its concept to arbitrary dimensions.**

4.1 Euclidean and unitary spaces

Z. Song

•Notion:

Unitary

space

Definition 4.1 Let $\mathcal{V}$ be n-dimensional linear space over $\mathcal{F}$, $(\boldsymbol{\alpha}, \boldsymbol{\beta}) \in \mathcal{V} \times \mathcal{V}$, $\langle \boldsymbol{\alpha}, \boldsymbol{\beta} \rangle \in \mathcal{F}$. The mapping $(\boldsymbol{\alpha}, \boldsymbol{\beta}) \mapsto \langle \boldsymbol{\alpha}, \boldsymbol{\beta} \rangle$ from $\mathcal{V} \times \mathcal{V}$ to $\mathcal{F}$ is said to be an inner product on $\mathcal{V}$ if it satisfies the following four properties for all $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$, $\boldsymbol{\gamma} \in \mathcal{V}$ and $k \in \mathcal{F}$:

1. $\langle \boldsymbol{\alpha}, \boldsymbol{\beta} \rangle = \overline{\langle \boldsymbol{\beta}, \boldsymbol{\alpha} \rangle}$, where $\bar{c}$ is the complex conjugate of c;

2. $\langle k\boldsymbol{\alpha}, \boldsymbol{\beta} \rangle = \bar{k} \langle \boldsymbol{\alpha}, \boldsymbol{\beta} \rangle$;

3. $\langle \boldsymbol{\alpha} + \boldsymbol{\beta}, \boldsymbol{\gamma} \rangle = \langle \boldsymbol{\alpha}, \boldsymbol{\gamma} \rangle + \langle \boldsymbol{\beta}, \boldsymbol{\gamma} \rangle$;

4. $\langle \boldsymbol{\alpha}, \boldsymbol{\alpha} \rangle \geq 0$. The equal sign holds if and only if $\boldsymbol{\alpha} = \mathbf{0}$.

A linear space together with the inner product $\langle \boldsymbol{\alpha}, \boldsymbol{\beta} \rangle : \mathcal{V} \times \mathcal{V} \mapsto \mathcal{F}$ is referred to as a unitary space.

4.1 Euclidean and unitary spaces

Z. Song

- Notion of Euclidean space

Definition 4.2 A linear space over $\mathcal{R}$ with an inner product is also called a Euclidean space.

- Properties of inner produce

1. $\langle \boldsymbol{\alpha}, k\boldsymbol{\beta} \rangle = k\langle \boldsymbol{\alpha}, \boldsymbol{\beta} \rangle$
2. $\langle \boldsymbol{\alpha}, \boldsymbol{\beta} + \boldsymbol{\gamma} \rangle = \langle \boldsymbol{\alpha}, \boldsymbol{\beta} \rangle + \langle \boldsymbol{\alpha}, \boldsymbol{\gamma} \rangle$
3. $\left\langle \sum_{i=1}^{s} k_i \boldsymbol{\alpha}_i, \boldsymbol{\beta} \right\rangle = \sum_{i=1}^{s} \bar{k}_i \langle \boldsymbol{\alpha}_i, \boldsymbol{\beta} \rangle$
4. $\left\langle \boldsymbol{\alpha}, \sum_{i=1}^{s} k_i \boldsymbol{\beta}_i \right\rangle = \sum_{i=1}^{s} k_i \langle \boldsymbol{\alpha}, \boldsymbol{\beta}_i \rangle$

4.1 Euclidean and unitary spaces

Z. Song

- Complex conjugate transpose of a matrix

Definition 4.2 Let $A \in \mathcal{F}^{m\times n}$ with the elements a_{ij}, $i = 1, 2, \ldots, m$, $j = 1, 2, \ldots, n$, $\bar{A} \in \mathcal{F}^{m\times n}$ be the matrix with the elements $\bar{a}_{ij}$. $A^* \triangleq \bar{A}^{\mathsf{T}}$ is referred to as the complex conjugate transpose of A.

- Properties of complex conjugate transpose of a matrix

1. $(A + B)^* = A^* + B^*$
2. $(kA)^* = \bar{k}A^*$
3. $(AB)^* = B^*A^*$
4. $(A^*)^* = A$
5. $(A^*)^{-1} = (A^{-1})^*$, if A is invertible.

- Example of unitary space, example 4.1-4.3

4.1 Euclidean and unitary spaces

- Characteristics in unitary space

Theorem 4.1 Let $\mathcal{V}$ be n-dimensional unitary space, $\{\boldsymbol{\alpha}_i, \quad i = 1, 2, \ldots, n\}$ be a basis of $\mathcal{V}$,

$$G \triangleq \begin{bmatrix} \langle \boldsymbol{\alpha}_1, \boldsymbol{\alpha}_1 \rangle & \langle \boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2 \rangle & \ldots & \langle \boldsymbol{\alpha}_1, \boldsymbol{\alpha}_n \rangle \\ \langle \boldsymbol{\alpha}_2, \boldsymbol{\alpha}_1 \rangle & \langle \boldsymbol{\alpha}_2, \boldsymbol{\alpha}_2 \rangle & \ldots & \langle \boldsymbol{\alpha}_2, \boldsymbol{\alpha}_n \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle \boldsymbol{\alpha}_n, \boldsymbol{\alpha}_1 \rangle & \langle \boldsymbol{\alpha}_n, \boldsymbol{\alpha}_2 \rangle & \ldots & \langle \boldsymbol{\alpha}_n, \boldsymbol{\alpha}_n \rangle \end{bmatrix}$$

be the Gramian of the basis vectors $\boldsymbol{\alpha}_i$, $i = 1, 2, \ldots, n$. Then for any two vectors

$$\boldsymbol{\alpha} = \sum_{i=1}^{n} \boldsymbol{\alpha}_i x_i = [\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \ldots \quad \boldsymbol{\alpha}_n]\, \boldsymbol{x},$$

$$\boldsymbol{\beta} = \sum_{j=1}^{n} \boldsymbol{\alpha}_j y_j = [\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \ldots \quad \boldsymbol{\alpha}_n]\, \boldsymbol{y}$$

in $\mathcal{V}$, $\langle \boldsymbol{\alpha}, \boldsymbol{\beta} \rangle = \boldsymbol{x}^* G \boldsymbol{y}$.

4.1 Euclidean and unitary spaces

- Gramian is Hermitian

Definition 4.3 Let $A \in \mathcal{F}^{n\times n}$. If $A^* = A$, A is said to be Hermitian; if $A^* = -A$, A is said to be skew-Hermitian.

When $\mathcal{F} = \mathcal{R}$, $A^* = A^{\mathsf{T}}$. $A^{\mathsf{T}} = A$ is said to be symmetric while $A^{\mathsf{T}} = -A$ is said to be skew-symmetric. From this definition, Gramian of a unitary space is Hermitian while Gramian of a Euclidean space is symmetric.

4.1 Euclidean and unitary spaces

- Relationship between Gramians of two deifferent bases-congruent

Theorem 4.2 Let $\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_n$ and $\boldsymbol{\alpha}'_1, \boldsymbol{\alpha}'_2, \ldots, \boldsymbol{\alpha}'_n$ be two bases of $\mathcal{V}$, G and G' be the corresponding Gramians, the transition matrix be P, i.e.

$$\left[\boldsymbol{\alpha}'_1 \quad \boldsymbol{\alpha}'_2 \quad \ldots \quad \boldsymbol{\alpha}'_n\right] = \left[\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \ldots \quad \boldsymbol{\alpha}_n\right] P$$

then the two Gramians G and G' are related by $G' = P^* G P$.

Definition 4.4 Let $A, B \in \mathcal{F}^{n\times n}$. If there exists a nonsingular matrix $P \in \mathcal{F}^{n\times n}$ such that $B = P^* A P$, then B is said to be congruent to A.

4.1 Euclidean and unitary spaces

- Metric in unitary spaces

Definition 4.5 Let $\mathcal{V}$ be a unitary space, the length (norm) of $\boldsymbol{\alpha} \in \mathcal{V}$ is defined as

$$\|\boldsymbol{\alpha}\| \triangleq \sqrt{\langle \boldsymbol{\alpha}, \boldsymbol{\alpha} \rangle}$$

Theorem 4.3 Let $\mathcal{V}$ be a unitary space over the field $\mathcal{F}$. Then the length function $\|\cdot\|$ has the following properties:

1. $\|\boldsymbol{\alpha}\| \geq 0$, the equal sign holds if and only if $\boldsymbol{\alpha} = \mathbf{0}$, (positive definiteness)

2. $\|k\boldsymbol{\alpha}\| = |k|\|\boldsymbol{\alpha}\|$, for any $k \in \mathcal{F}$,

3. $\|\boldsymbol{\alpha} + \boldsymbol{\beta}\| \leq \|\boldsymbol{\alpha}\| + \|\boldsymbol{\beta}\|$ (triangular inequality)

4. $|\langle \boldsymbol{\alpha}, \boldsymbol{\beta} \rangle| \leq \|\boldsymbol{\alpha}\| \cdot \|\boldsymbol{\beta}\|$ (Cauchy-Schwarz inequality)

4.2 Orthonormal basis and the Gram-Smith process

Z. Song

•Concept of orthonormal basis

Definition 4.6 Two vectors $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ in a unitary space $\mathcal{V}$ are said to be orthogonal if $\langle \boldsymbol{\alpha}, \boldsymbol{\beta} \rangle = 0$, denoted $\boldsymbol{\alpha} \perp \boldsymbol{\beta}$. The set $\{\boldsymbol{\alpha}_i,\ i = 1, 2, \ldots, r\} \subset \mathcal{V}$ is referred to as an orthogonal system if it does not contain the zero vector and the elements $\boldsymbol{\alpha}_i$, $i = 1, 2, \ldots$, are pairwise orthogonal. An orthonormal system is an orthogonal system with all the vectors being of unit length.

From the definition the following are obvious:
1.The set $\{\boldsymbol{\alpha}_i,\ i = 1, 2, \ldots, r\}$ is an orthogonal system if and only if

$$\langle \boldsymbol{\alpha}_k, \boldsymbol{\alpha}_j \rangle = 0 \qquad \forall k \neq j$$

2. The set $\{\boldsymbol{\alpha}_i,\ i = 1, 2, \ldots, r\}$ is an orthonormal systems if and only if

$$\langle \boldsymbol{\alpha}_k, \boldsymbol{\alpha}_j \rangle = \delta_{kj} = \begin{cases} 1 & \text{if } k = j \\ 0 & \text{otherwise} \end{cases}$$

3. zero vector is orthogonal to every vector. Conversely, the vector that is orthogonal to every vector is the zero vector.

4.2 Orthonormal basis and the Gram-Smith process

Z. Song

- Concept of orthonormal basis

Theorem 4.4 The vectors $\alpha_1, \alpha_2, \ldots, \alpha_r$ of an orthogonal system are linearly independent.

Definition 4.7 Let $\mathcal{V}$ be n-dimensional unitary space. An orthogonal system of $\mathcal{V}$ consisting of n vectors is called an orthogonal basis, an orthonormal system of $\mathcal{V}$ consisting of n vectors is called an orthonormal basis.

4.2 Orthonormal basis and the Gram-Smith process

Z. Song

- Concept of orthonormal basis

Corollary 4.1 Let $\mathcal{V}$ be an n-dimensional unitary space, $\{\boldsymbol{\alpha}_i, \quad i = 1, 2, \ldots, n\}$ be an orthonormal basis of $\mathcal{V}$. Then for any two vectors

$$\boldsymbol{\alpha} = \sum_{i=1}^{n} \boldsymbol{\alpha}_i x_i = [\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \ldots \quad \boldsymbol{\alpha}_n]\, \boldsymbol{x},$$

$$\boldsymbol{\beta} = \sum_{j=1}^{n} \boldsymbol{\alpha}_j y_j = [\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \ldots \quad \boldsymbol{\alpha}_n]\, \boldsymbol{y}$$

in $\mathcal{V}$,

$$\langle \boldsymbol{\alpha}, \boldsymbol{\beta} \rangle = \boldsymbol{x}^* \boldsymbol{y} = \langle \boldsymbol{x}, \boldsymbol{y} \rangle. \qquad (1)$$

4.2 Orthonormal basis and the Gram-Smith process

Z. Song

- the
- Gram
- -Smith
- process

Let $\alpha_1, \alpha_2, \ldots, \alpha_r$ be vectors in an n-dimensional unitary space. The Gram-Schmidt orthonormalization process can be accomplished through two steps.

Step 1: Orthogonalization of the vectors. Let

$$\begin{aligned}
\beta_1 &= \alpha_1 \\
\beta_2 &= \alpha_2 - \frac{\langle \beta_1 , \alpha_2 \rangle}{\langle \beta_1 , \beta_1 \rangle}\beta_1 \\
\beta_3 &= \alpha_3 - \frac{\langle \beta_1 , \alpha_3 \rangle}{\langle \beta_1 , \beta_1 \rangle}\beta_1 - \frac{\langle \beta_2 , \alpha_3 \rangle}{\langle \beta_2 , \beta_2 \rangle}\beta_2 \\
&\vdots \\
\beta_r &= \alpha_r - \frac{\langle \beta_1 , \alpha_r \rangle}{\langle \beta_1 , \beta_1 \rangle}\beta_1 - \frac{\langle \beta_2 , \alpha_r \rangle}{\langle \beta_2 , \beta_2 \rangle}\beta_2 - \cdots - \frac{\langle \beta_{r-1} , \alpha_r \rangle}{\langle \beta_{r-1} , \beta_{r-1} \rangle}\beta_{r-1}
\end{aligned}$$

It can be verified that the vectors $\beta_1, \beta_2, \ldots, \beta_r$ are pairwise orthogonal.

Step 2: Normalization of the vectors obtained in step 1. This can be accomplished simply by letting

$$\gamma_1 = \frac{\beta_1}{\|\beta_1\|}, \quad \gamma_2 = \frac{\beta_2}{\|\beta_2\|}, \quad \ldots, \quad \gamma_r = \frac{\beta_r}{\|\beta_r\|}.$$

Thereom 4.5 The vectors γ_i, $i = 1, 2, \ldots, r$ constitutes an orthonormal system if and only if the vectors $\alpha_1, \alpha_2, \ldots, \alpha_r$ are linearly independent.

4.3 Unitary transformations

Z. Song

- Notion of unitary transformation and orthogonal transformation

Definition 4.8 Let $\mathcal{V}$ be n-dimensional unitary space, $\mathscr{U}$ is a transformation on $\mathcal{V}$. $\mathscr{U}$ is referred to as a unitary transformation if $\langle \mathscr{U}(\alpha), \mathscr{U}(\beta) \rangle = \langle \alpha, \beta \rangle \ \forall \alpha, \beta \in \mathcal{V}$.

•Unitary transformation on a Euclidean space is called orthogonal transformation.

Thereom 4.6 A unitary transformation $\mathscr{U}$ is a linear transformation.

4.3 Unitary transformations

Z. Song

- Matrix representation of unitary transformation is unitary matrix

  - Notion of unitary matrix

Definition 4.9 An $n \times n$ matrix A is referred to as unitary if

$$A^* A = A A^* = E\,.$$

The set of all unitary matrices is denoted $\mathcal{U}^{n\times n}$. An $n \times n$ real matrix A is said to be orthogonal if

$$A^{\mathsf{T}} A = A A^{\mathsf{T}} = E\,.$$

The set of all $n \times n$ orthogonal matrices is denoted $\mathcal{E}^{n\times n}$.

Thereom 4.7 An $n \times n$ matrix A is unitary if and only if its n column (row) vectors form an orthonormal basis for $\mathcal{C}^n$.

4.3 Unitary transformations

Z. Song

- Matrix representation of unitary transformation is unitary matrix-continued

Definition 4.8 Let $\mathscr{U} \in \mathcal{L}(\mathcal{V})$. The following statements are equivalent

1. $\mathscr{U}$ is unitary;
2. $\|\mathscr{U}(\boldsymbol{\alpha})\| = \|\boldsymbol{\alpha}\| \ \forall \boldsymbol{\alpha} \in \mathcal{V}$;
3. $\mathscr{U}$ maps an orthonormal basis of $\mathcal{V}$ to an orthonormal basis;
4. the representation of $\mathscr{U}$ with respect to an orthonormal basis is a unitary matrix.

4.3 Unitary transformations

- Matrix representation of unitary transformation is unitary matrix-

continued

Corollary 4.2 If $A,\ B \in \mathcal{U}^{n\times n}$ $(\mathcal{E}^{n\times n})$, then

1. $A^{-1} = A^* \in \mathcal{U}^{n\times n}$ $(A^{-1} = A^{\mathsf{T}} \in \mathcal{E}^{n\times n})$;

2. $|\det A| = 1$ $(\det A = \pm 1)$;

3. $A^{\mathsf{T}} \in \mathcal{U}^{n\times n}$;

4. $AB,\ BA \in \mathcal{U}^{n\times n}$ $(\mathcal{E}^{n\times n})$.

- Example 4.7 , 4.8